EEES 302 - Digital Design
Experiment - 2

1. Introduction:

Combinational logic circuits are crucial for industrial automation, such as controlling conveyer belts.

To design these circuits efficiently, engineers use Karnaugh Maps to visually simplify Boolean expressions, minimizing the number of logic gates required and reducing hardware costs.

2. Purpose:

The primary objective of this experiment is to design, build and test a combinational logic circuit that governs the stopping mechenism of a conveyer belt carrier.

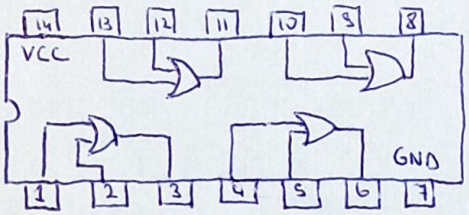
The system must trigger the alarm under two specific conditions.

A) If there is no ink and bottle in the same time.

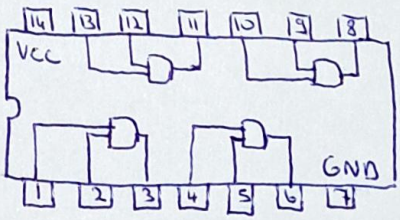
B) When the user push the emergency button.

The experiment aims to portically apply a 3-variable truth table, simplify the resulting logic using a Karnaugh Map and physically construct the minimized Boolean function.

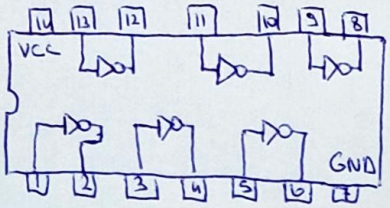
3. Equipment List:



74LS32
TTL OR Gate IC



74LS08
TTL AND Gate IC



74LS04 TTL NOT Gate IC

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EEES 302 - Digital Design
Experiment 2

4. Preliminary Work:

An alarm is to be used in automated ink bottling plant. A conveyer belt carries the empty ink bottles past the filling spotit. The alarm is to sound if any of the following conditions occur:

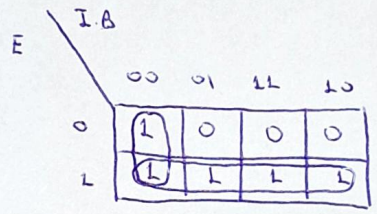
A) If there is no ink and bottle in the same time.
B) When the user push the emergency button.

I = Ink in the tank B = Bottles on the conveyer belt C = Emergency button

Karnough Map:

E	I	B	Alarm
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

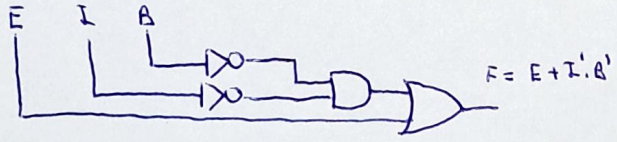
Truth Table



$F = \bar{E} + I'.B'$

5. Experimental Procedure:

The logical expression is simplified by using k-map. Then the circuit is built by using IC's.



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EEES 302 - Digital Design
Experiment - 2

The circuit is shown tested by using DMM, ossilator. The truth table conditions are implemented as order.

The implemented circuit gives the output 1 for the all inputs 1 of E.

The output is 1 for the both input I and B are 0.

6. Result:

To evaluate the system, we established a truth table mapping the three inputs (E, I, B) to the output F.

The physical circuit was built using standart logic IC's. 74LS32 is used for OR gate. 74LS08 for AND gate. 74LS04 for NOT gate.

The system successfully registered a "High" output anytime the emergency input was 1.
The system gived the output 1 when the system was running completely empty.

7. Conclusion:

The experiment demonstrated how combinational logic and Karnaugh maps are used to design different condition dependent safety systems.

By mapping the physical conditions into a truth table, we derived and built the minimum-hardware Boolean function.

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